

48V Electrical Architecture

Document role

- This file owns the final 48V electrical architecture for the active build baseline.
- Use it for house-bank structure, alternator-path direction, shutdown logic, and manual control-path intent.
- Keep full conductor schedules, fuse matrices, and layout-level implementation detail in docs/implementation/ .
- Keep unresolved vendor gates, risk state, and follow-up closure items in docs/core/TRACKING.md .
- Keep broad project sequencing and day-to-day execution framing in docs/core/PROJECT.md or the active plan docs.

As-of date: 2026-03-20

Purpose: hold the finalized, concise 48V house and alternator architecture in one place so wiring, protection, shutdown behavior, and BOM references are easy to understand without re-reading the historical trade studies.

Related docs:

- [Electrical topology diagram](#)
- [Electrical fuse schedule](#)
- [Systems](#)
- [Tracking](#)
- [Estimated BOM](#)

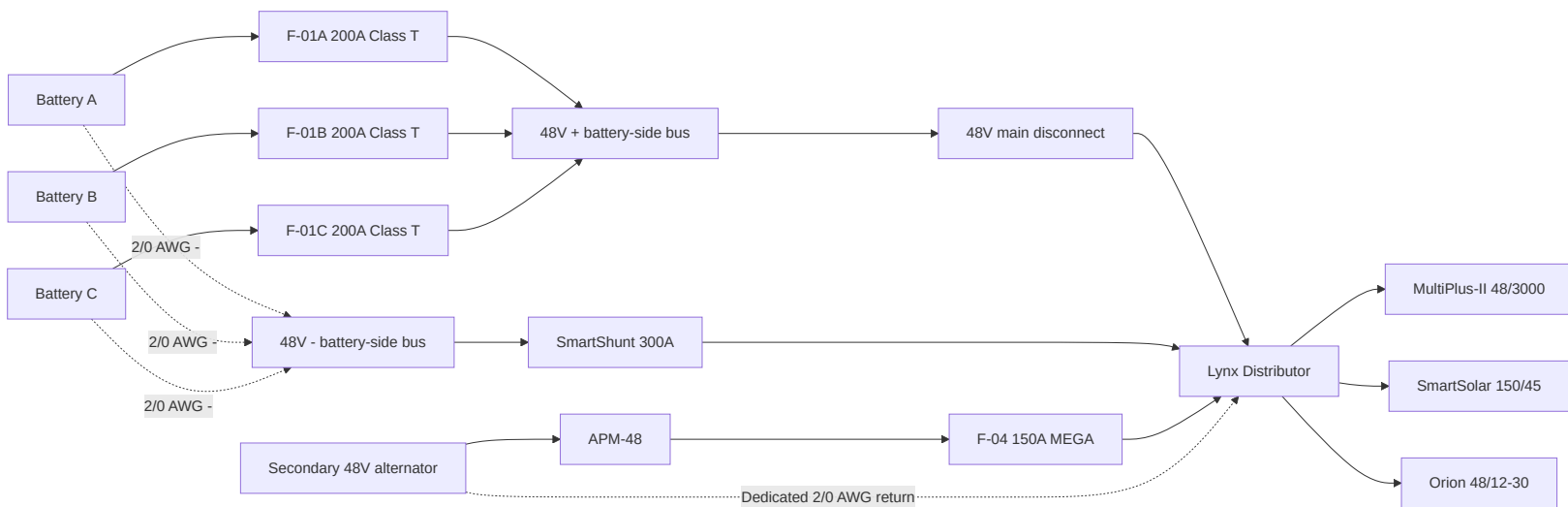
Final direction

- House architecture stays 48V core with 3x 51.2V 100Ah batteries in parallel.
- Alternator charging is the dedicated 48V secondary alternator path, not Sterling B2B.
- Protection baseline is Mechman + WS500 + Balmar APM-48 .
- Manual alternator-charge enable/disable is through Ford Upfitter Switch #3 .
- Upfitter #3 is a low-current control signal only. It does not carry alternator output current.
- WS500 white Feature-In is reserved for future fault-interlock work, not required in Phase 1.

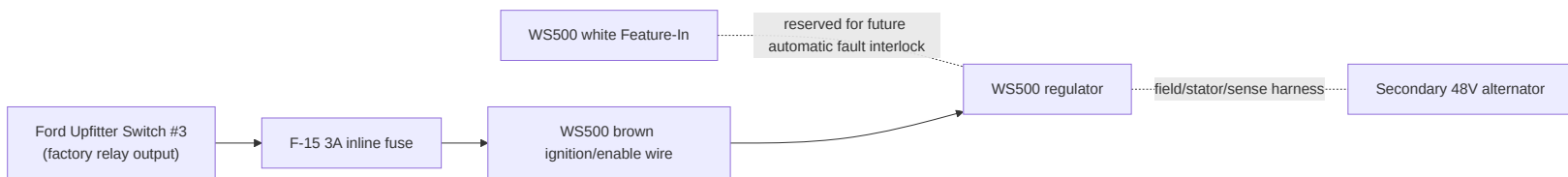
Locked component set

Function	Locked baseline	BOM row(s)
House batteries	3x Dumfume 51.2V 100Ah LiFePO4	3
Main house disconnect	Victron 275A battery switch	5
Main fused distribution	Victron Lynx Distributor M10	6
Battery branch protection	F-01A/B/C 200A Class T (provisional pending final battery limit confirmation); owner confirmed 3 holders and 4 slow-blow Class T fuses total	7
Inverter/charger	MultiPlus-II 48/3000/35-50	12
48V to 12V charger	Orion-Tr Smart 48/12-30	20
Monitoring	Cerbo GX + SmartShunt 300A	22, 23
Alternator kit	Mechman 48v secondary alternator kit with WS500	168
Load-dump clamp	Balmar APM-48	169
Alternator branch fuse	F-04 150A MEGA (58V/80V) in Lynx Slot 3	170
WS500 low-current fuse set	F-12, F-13, F-14	171
WS500 Upfitter #3 control kit	F-15 + control wire + holder/terminals	176

48V power path



Alternator control path



Control logic

- Upfitter #3 ON : WS500 brown wire is energized, regulator is allowed to run, alternator charging can occur.
- Upfitter #3 OFF : WS500 is disabled through the brown wire, regulator field output collapses, and alternator charging stops.
- Use Upfitter #3 as the manual alternator-charge shutdown control.
- Do not use the main 48V disconnect as the first alternator shutdown method while the engine is running and the secondary alternator is charging.

Why Upfitter #3

- It is already a factory-switched, relay-driven 12V control source.
- It keeps the WS500 control in the truck cab without adding a separate aftermarket switch.
- It is suitable for a low-current enable signal, but not for any high-current alternator or battery conductor.
- A local inline fuse is still required because the factory upfitter circuit fuse is much larger than the small-gauge WS500 control wire.

Fusing and wire baseline

ID	Function	Locked value	Wire basis
F-01A/B/C	Battery branch positive protection	200A Class T provisional	2/0 AWG
F-04	Alternator branch into Lynx Slot 3	150A MEGA (58V/80V)	2/0 AWG
F-12	WS500 power lead	10A baseline (15A if required by alternator case)	harness lead
F-13	WS500 battery positive sense	3A; fuse/holder voltage class must be verified for the 48v bank or proven by WS500 harness documentation	harness lead
F-14	WS500 current-sense lead	5A where required; confirm actual voltage/reference from WS500 documentation before hardware lock	harness lead

F-15	Upfitter #3 to WS500 brown ignition wire	3A inline ATO/ATC	16 AWG TXL/GXL control wire
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Major conductors

- Battery branch and main 48V trunk wiring: 2/0 AWG .
- Secondary alternator positive path (ALT B+ -> APM-48 -> F-04 -> Lynx): 2/0 AWG , ~20 ft one-way planning basis.
- Secondary alternator dedicated negative return (ALT B- -> Lynx -): 2/0 AWG , ~20 ft one-way planning basis.
- Orion 48V feeder and MPPT battery leads: 6 AWG .
- Upfitter #3 control lead to WS500 brown wire: 16 AWG TXL/GXL planning basis, ~6 ft one-way assumed until measured.

APM-48 wiring intent

- Mount the APM-48 at the alternator end of the 48V branch, as close to the alternator output as practical.
- Connect APM red to alternator B+ .
- Connect APM black to alternator B- , or to the approved alternator ground/case point if the alternator is not isolated-ground.
- Do not stack the APM ring terminals under the main battery cable lugs unless the product instructions for the exact unit explicitly allow it.

Normal operating sequence

1. Main 48V disconnect closed.
2. Upfitter #3 switched ON .
3. Engine running.
4. WS500 enabled and regulating.
5. Alternator charges the house bank through APM-48 and F-04 .

Fault and shutdown sequence

If a battery pack trips or an alternator fault is suspected:

1. Switch Upfitter #3 OFF to disable the WS500 .
2. Wait for alternator charge current to collapse.
3. Open the main 48V disconnect only after alternator charging is no longer active, if full house shutdown is required.

Reason:

- The regulator should be shut down first.
- The APM-48 is a protection layer, not the primary shutdown method.

One-battery-trip behavior

- If one battery BMS disconnects but the other two remain online, the 48V bus should stay up.
- The system does not immediately become a full-bank load dump event just because one battery disappears.
- The practical effect is that charge and discharge current redistribute to the remaining batteries.
- The real hazard is a cascade where the second and third battery also disconnect under active alternator charge.

What is still not fully closed

- Final Mechman kit fitment/content confirmation for the exact truck.
- Final WS500 harness polarity confirmation (PH vs NH).
- Official vendor confirmation that the documented Dumfume battery/BMS behavior is acceptable with the WS500 .
- Exact alternator negative/case isolation behavior in the installed Mechman kit.
- Whether future automatic fault-interlock logic should be added on the WS500 Feature-In or through an external relay.

Rule for future edits

- Update this file first for any 48V architecture, alternator-control, or shutdown-strategy change.
- Keep the implementation files for wiring detail and fuse placement, not for high-level architecture decisions.